Subject :-Introduction to Cyber Security

Unit -1 Introduction to Cyber Security

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- Defining Cyberspace
- Overview of Computer and Web-technology
- Architecture of cyberspace

- Communication and web technology
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- Regulation of cyberspace
- Concept of cyber security

Cyberspace

- The term “cyberspace” was introduced in the early 1980s by William Gibson, an American science fiction novelist in his work Neuromancer.

- It refers to the virtual environment created by interlocking networks of

computers and electronic systems.

- Cyberspace is that realm in which online communication, data exchange, and digital interactions take place.

Cyberspace

• Cyberspace refers to the virtual digital environment created by interconnected computers, networks, devices, and data. • It is not a physical space, but a digital world where:

- Communication happens
- Data is stored and shared
- Services and applications operate

• **Examples of cyberspace:**

- Social media platforms (Instagram, Facebook)
- Email systems
- Online banking portals
- Cloud services
- E-commerce sites (Amazon, Flipkart)

Overview of Computer and Web-technology

- **Computer Technology**
- Components of a Computer System
    a) Hardware
        a) Input devices
        b) Output devices
        c) Storage
        d) Processor

b) Software

a) System Software

b) Application Software

c) Networks

d) Data storage and Processing

Web Technology

- Web technologies enable websites, online services, and applications to function.

1. Web Browsers
2. Web servers
3. Web Pages & Languages
4. Protocols

Architecture of Cyberspace

- Cyberspace has multiple layers and components working together.
- **Layers of Cyberspace:**
    - Physical Layer
    - Network Layer
    - Application Layer
    - Content(Data) Layer:
    - User Layer:

Communication and Web Technology

- Cyberspace works only because **communication technologies** and **web technologies** connect devices, transfer data, and deliver online services. Both are the building blocks of how the internet functions.

- Communication technology refers to the **methods, devices, and systems** used to send data from one point to another over a network or the internet.(How data travels).

- Web Technology includes **tools, technologies, and protocols** that allow the creation and functioning of websites and web applications. (How websites and applications work).

Internet

- The Internet is a global network of millions of computers and devices that are connected together to share information and communicate with each other.

- It is also called a network of networks.

- Definition: Internet is a worldwide system that connects computers so people can send information, access websites, communicate, and use online services from anywhere in the world.

Features of Internet

• Global Connectivity •

Uses Standard Protocols •

Decentralized Network •

Provides Multiple Services

World Wide Web (WWW)

- The World Wide Web (WWW or Web) is a collection of websites and web pages that you can access through the internet using a browser.

- The Web is a system of web pages linked together using hyperlinks and accessed through the internet.

- It is one of the major services provided on the internet (just like email, chat, cloud storage).

WWW

1. It is made of **Web Pages**

2. Accessed Using **Web Browsers** 3. It Works on

Client–Server Model 4. Uses **HTTP/HTTPS** Protocols. 5.

Uses **URLs** to Locate Websites

Internet infrastructure for data transfer and governance

Infrastructure for Data Transfer (How data moves)

- End-User Devices
- Access Networks (First point of connection)
- ISPs (Internet Service Providers)
- Routers, Switches, and Gateways
- Data Centers & Web Servers
- Submarine Fiber-Optic Cables
- Internet Exchange Points (IXPs)

- CDNs (Content Delivery Networks)
- DNS (Domain Name System)

Internet Governance

- ICANN (Internet Corporation for Assigned Names and Numbers)
- IANA (Internet Assigned Numbers Authority)
- IETF (Internet Engineering Task Force)
- W3C (World Wide Web Consortium)
- National Governments
- ISPs and Telecom Companies

Internet Society (ISOC)

- The **Internet Society (ISOC)** is a **global, non-profit organization** founded in **1992** to ensure the **open development, evolution, and use of the Internet** for the benefit of all people across the world.

- A professional membership society that promotes the use and future development of the Internet.

Main Purpose of Internet Society

1. Supports Internet Standards
   - ISOC is the organizational home for:
     1. **IETF (Internet Engineering Task Force)**
     2. **IAB (Internet Architecture Board)**
2. Promotes Internet Access for All

3. Works for Internet Security
4. Advocates for Policy and Governance
5. Education & Awareness

Cybersecurity

- **Cyber Security** is the practice of **protecting computers, networks, data, and digital systems** from unauthorized access, attacks, damage, or misuse.

- Cybersecurity is the combination of methods, processes, tools, and behaviors that protect computer systems, networks, and data from **cyberattacks** and unauthorized access.

- Cybersecurity is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks. It's also known as information technology security or electronic information security.

Why Is Cybersecurity Important

- Protection of Sensitive Data
- Business Continuity and Reputation
- Economic and Regulatory Implications •

National Security and Critical Infrastructure •

Trust and Reputation

Shalini Chandran new ad for RBI,cyber crime awareness

https://www.youtube.com/watch?v=Sgs4K2eAZhg

CIA Triad


CIA Triad
C
CONFIDENTIALITY
Confidentiality means the prevention of unauthorized disclosure of information.
01
I
INTEGRITY
Integrity means maintaining and assuring that data cannot be modified in an unauthorized or undetected manner.
02
A
AVAILABILITY
Availability means information should be readily accessible for the authorized users
03

CIA Triad

- **Confidentiality-** Protecting information from unauthorized access. • **Integrity-** Ensuring data is not modified or altered.
- **Availability -** Ensuring systems and information are available when needed.

Types or Domains of Cybersecurity



Types or Domains of Cybersecurity

- **Network Security -** Protects **networks and data during transfer** from unauthorized access or attacks.
- **Information / Data Security-** Protects **data in all forms**—digital or physical.
- **Endpoint Security -**Secures individual devices
- **Application Security-** Secures software applications from vulnerabilities.
- **Cloud Security-** Secures data, applications, and infrastructure in cloud environments.
- **Operational Security (OpSec)-** Focuses on processes, policies, and procedures for secure operations.
- **Internet of Things (IoT) Security-** IoT security focuses on safeguarding interconnected devices and networks within the IoT ecosystem.

Cyber Threat

- A cyber threat is the possibility that someone may use digital techniques to damage systems, steal data, or disrupt online services.

Common Types of Cyber Threats

1. Malware
2. Phishing
3. Ransomware
4. Denial-of-Service (DoS/DDoS) Attacks
5. Social Engineering
6. Insider Threat
7. Zero-Day Threat
8. MITM (Man-in-the-Middle) Attack

Cyber attack

- A cyber attack is an intentional action taken by a hacker to break into a system and cause harm.

Common Types of Cyber Attack

1. Malware Attack
2. Ransomware Attack
3. Phishing Attack
4. DDoS Attack
5. MITM Attack
6. SQL Injection
7. Brute Force Attack
8. Data Breach

Issues and challenges of cyber security.

- Rapidly Evolving Threats
- Lack of Awareness Among Users
- Cloud Security Issues
- Rapid Growth of Internet of Things (IoT)
- Increasing Number of Cyber Attacks
- Regulatory Compliance

THANK YOU